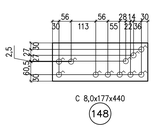
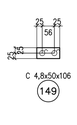
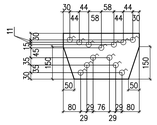
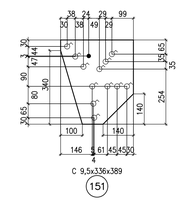
Eletrobras
APROVADO
Por: Wallace Castro em 30/07/2026

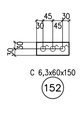
PARAFUSOS e ACESSÓRIOS	QUANTIDADES PARA 1 (UMA) TORRE	
PARAFUSOS / COMPRIMENTOS	M12	M16
30	52	–
35	81	–
40	32	98
45	–	70
50	–	16
55	–	–
60	–	–
65	–	–
70	–	–
75	–	–
ARRUELA PLANA ESP. 3,0	251	254
"PALNUT"	165	184
ESPESSURAS		
# 3,0	–	–
# 4,0	6	–
# 5,0	52	–
# 6,0	–	–
# 7,0	2	–
# 8,0	–	–
PARAFUSO DEGRAU PD-18M	20	

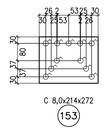


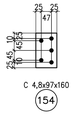


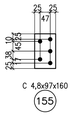


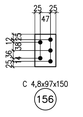


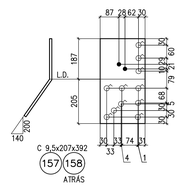


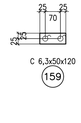


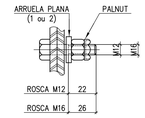


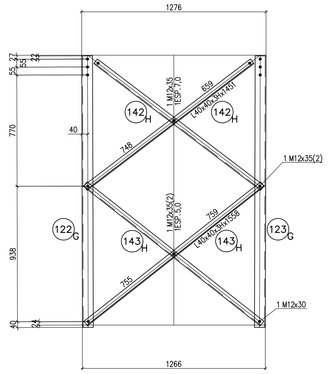


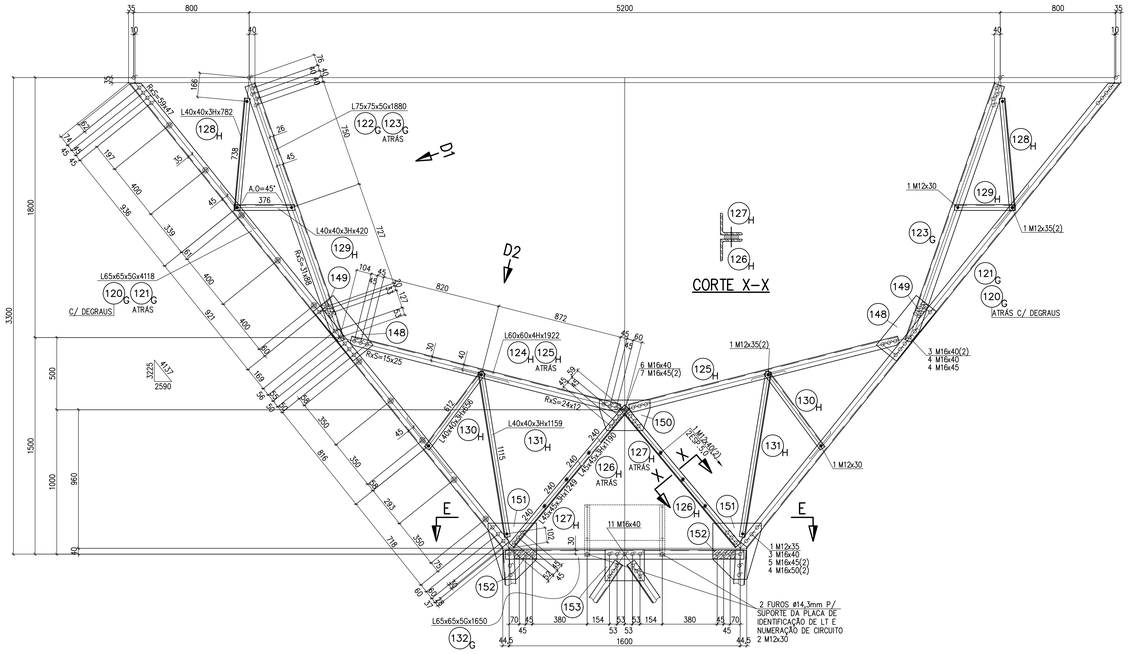


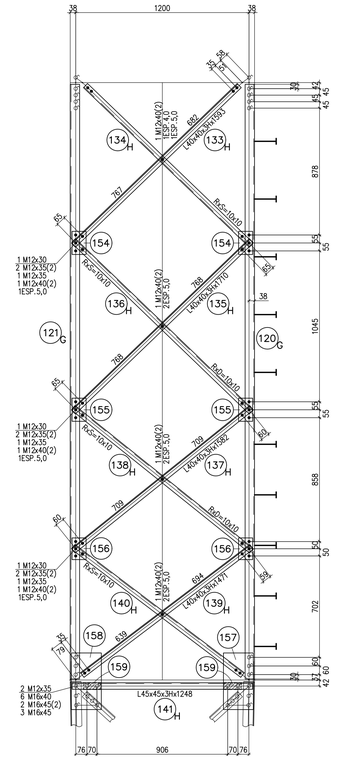


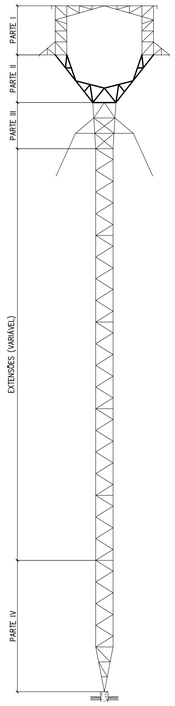


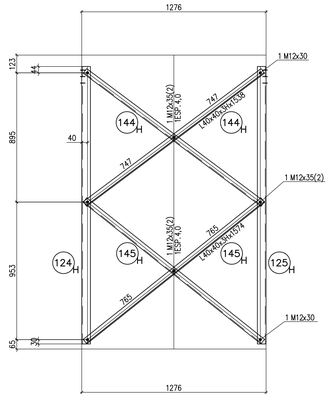


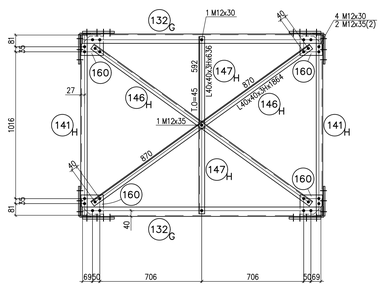


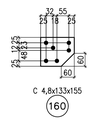












REV.	ALTERAÇÃO	DATA	DES.	VERIF.	APROV.
00	APROVADO	13/07/2026	L.PENIDO	M.D.R	H.D.G.
00A	EMISSÃO INICIAL	05/03/2026	F.A.F.	L.C.C.	H.D.G.

POS. 120 : 160

FINALIDADE

PARA APROVAÇÃO

SIMBOLOGIA DOS PARAFUSOS

- Parafusos M12 – Furos ø13,5 – A=22
- Parafusos M16 – Furos ø17,5 – A=25
- Parafusos M20 – Furos ø21,5 – A=30
- Parafusos M24 – Furos ø26,0 – A=40
- Parafusos Degraus PD16 – Furos ø17,5
- Furos Conforme ø indicado

NOMENCLATURA DE RECORTES

RxS (Simples) · A.0 45° (Simples Aba Oposta) · T.0=A (Total Aba Oposta) · RxD (Duplo) · D.0+45° (Duplo Aba Oposta) · C/L L1.0=C+B (Intermed. Aba Oposta)

TORQUE PARA APERTO DOS PARAFUSOS

ø	Min. (kg.m)	Max. (kg.m)
M12	3,5	4,5
M16	7,5	11,5
M20	13,0	22,0

REFERÊNCIAS

BNEL_ADR2451-D1002 ESPECIFICAÇÕES e NOTAS GERAIS
BNEL_ADR2451-D1002 TORRE BÁSICA: PARTE I
BNEL_ADR2451-D1004 TORRE BÁSICA: PARTE III
BNEL_ADR2451-D1005 TORRE BÁSICA: PARTE IV

HENRIQUE GONTIJO
RESPONSÁVEL TÉCNICO
CREA 108382/MG

LOTE 05 - ANEEL 01/2024

Nº DOCUMENTO: LTRH1.06-230-PE-ME.DMT-0002

Nº EMPREENDIMENTO: NOVA ERA INTEGRAÇÃO
REVISÃO: 00
UNIDADE: LOTE 5 - LT 230 kV
FOLHA: 01/01
LOCAL: BOM NOME - BOM NOME II C1
Nº CONTRATO: -
CÓDIGO INSTALAÇÃO: -
Nº LEGADO: -
TAG: -

TÍTULO: TORRE DE SUSPENSÃO ESTAIADA DA ESTRUTURA TÍPICA BNEL
TORRE BÁSICA: PARTE II

Nº CR GONTIJO: BNEL_ADR2451-D1003
ESCALA: 1/10 - 1/15
REVISÃO: R- 00

Formato A0 (1189x841mm)